

Stochastic TTA & Forward

1. Multi-Scale Ingestion

Input Frame (I)



0.85x



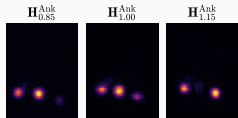
1.00x



1.15x

...

CNN / ViT Backbone
(Frozen HRNet / ViTPose)



$s \in \{0.85, \dots, 1.15\} \times \{\text{Orig, Flip}\}$

Continuous Aggregation

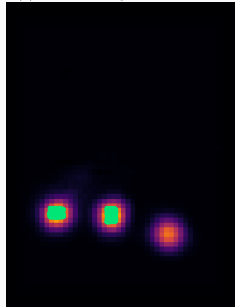
2. Sharpness-Weighted MC

$$\mathbf{P}_k(\mathbf{x}) = \sum_{n=1}^N w_k^{(n)} \tilde{\mathbf{H}}_k^{(n)}(\mathbf{x}), \quad w_k^{(n)} = \frac{C_k^{(n)}}{\sum_m C_k^{(m)}}$$

$$C_k^{(n)} = (\rho_k^{(n)})^\alpha \cdot \left[1 - \frac{1}{1 + \frac{\rho_k^{(n)}}{(\rho_k^{(n)} + \epsilon)}} \right]^\beta$$

Dequant MC: $\tilde{\mathbf{x}}_m = \mathbf{x}_m + \epsilon, \quad \epsilon \sim \mathcal{U}(-0.5, 0.5)$

$\mathbf{P}_{\text{R_Ankle}}(\mathbf{x}) + \text{MC Samples } (N=1000, T=0.3)$

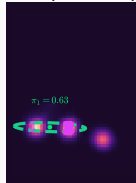


◦ Green circles: Sub-pixel MC Samples $\tilde{\mathbf{x}}_m \sim \mathbf{P}_k$

Probabilistic Mixture

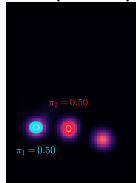
3. EM Fitting + BIC

$K=1$ (Unimodal)



$\text{BIC}_1 = 8604.7 \mid \pi_u = 37.0\%$

$K=2$ (Bimodal)



$\text{BIC}_2 = 5739.9 \mid \pi_u = 0.0\%$

$\text{BIC}_2 < \text{BIC}_1$
 $\Rightarrow$ Bimodal R_Ankle

R_Ankle Uncertainty

1. GMM Modes:

$$\{\mu_1, \Sigma_1, \pi_1\}$$

$$\{\mu_2, \Sigma_2, \pi_2\}$$

2. Spatial Metric:

$$|\Sigma_{\text{final}}| = 0.077$$

$$U_{\text{gmm}} = 0.008$$

3. Softmax Fusion:

$$U_{\text{base}} = 0.33$$

$$U_{\text{adapt}} = 0.19$$

4. OoD Metric:

$$\pi_{\text{uniform}} = 0.000$$

Kinematic MRF Graph

4. Topology Decoding

Global Tree Optimization:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \sum_u \phi_u(x_u) + \sum_{(u,v)} \psi_{uv}(x_u, x_v)$$

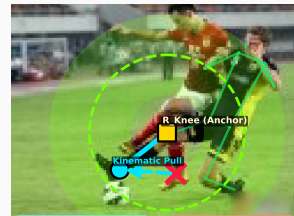
Unary Cost: $\phi_u(x_u) = -\ln \mathbf{P}_u(x_u)$

$$\text{Kinematic Pairwise: } \psi_{uv} = \frac{(\|\mathbf{x}_u - \mathbf{x}_v\| - \mu_{uv})^2}{2\sigma_{uv}^2}$$

Prior (R_Knee $\rightarrow$ R_Ankle):

$$\mu_{uv} = 33.7\text{px}, \quad \sigma_{uv} = 17.6\text{px} \quad (\sigma_{uv}^2 = 310.1\text{px}^2)$$

Kinematic Prior Band & Swap Correction



MRF Solution $\mathbf{x}^* (\mu_1)$
 $d = 26.1\text{px} \Rightarrow (d - \mu)^2 = 57.9\text{px}^2$
[Selected $\checkmark$ | In Prior Band]

GMM Unary μ_2 (Swap)
 $d = 18.3\text{px} \Rightarrow (d - \mu)^2 = 235.7\text{px}^2$
[Pruned X | High Penalty]